Class Imbalance in Histopathology Patch Classification

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1 Motivation

Histopathology datasets are long-tailed almost by construction. Common cancers and benign tissue dominate archives, while rare subtypes and atypical lesions, often diagnostically decisive, contribute few patients and patches. In convolutional neural networks, imbalance biases predictions toward frequent classes and lowers minority recall (Buda et al., 2018). In medical imaging, it also distorts calibration, and conclusions depend on whether evaluation uses balanced metrics (Mosquera et al., 2024). No single remedy works across long-tailed medical benchmarks (Ju et al., 2024), and researchers continue to propose histopathology-specific models (Cong et al., 2022). Pathology foundation models such as UNI and Virchow2 are commonly evaluated with linear classifiers trained on their frozen patch features (Chen et al., 2024; Zimmermann et al., 2024). Yet how much class imbalance costs such classifiers, and why, remains insufficiently measured in isolation.

This summary follows three questions. How much do discrimination and calibration degrade as the training class distribution becomes more skewed (Section 3)? What causes this damage, and why are some classes and tasks more vulnerable than others (Section 4)? How much of the damage can mitigation methods recover?

2 Setup

Multinomial logistic regression on Virchow2 patch features (Zimmermann et al., 2024) addresses two tasks: TCGA-UT, 30 cancer types (Komura et al., 2022), and BRACS, 7 breast-lesion subtypes from normal tissue to invasive carcinoma (Brancati et al., 2022). Allocation conditions use the same patients per class (20 on TCGA-UT, 10 on BRACS) and total patch budget; only class shares change. An exponential profile sets the largest-to-smallest class ratio ρ from 1 to 100 (Figure 1). Class-to-rank assignment is randomized over 30 draws (3 splits $\times$ 10 draws); a native condition preserves each dataset's class shares. The outcome is *damage*, $\Delta(\rho) = \mathrm{BA}(1) - \mathrm{BA}(\rho)$: the loss in test patient-macro balanced accuracy (BA) relative to the balanced condition, with paired bootstrap 95% intervals. Damage counts as material from 1 percentage point (pp).

The encoder stays frozen when measuring damage and its causes. Mitigation methods that adapt the encoder fine-tune it through low-rank adaptation (LoRA) (Hu et al., 2022). To rule out dependence on one encoder, the conditions at $\rho = 10$ and $\rho = 100$ are repeated on the same patients and patches with UNI2-h, the successor of UNI (Chen et al., 2024).

3 Effects of class imbalance

Balanced accuracy in the balanced condition reaches 72.1% on TCGA-UT and 39.6% on BRACS. Published evaluations show the same broad gap: region-of-interest (ROI)-level linear probes score 60–68% on TCGA-UT (Mahmood Lab, 2025), while slide-level linear probes score 34.8–48.1% on BRACS (Vaidya et al., 2025). TCGA-UT spans cancer types across organs, while BRACS separates breast lesions with similar morphology, such as atypical ductal hyperplasia (ADH) and ductal carcinoma in situ (DCIS) (Brancati et al., 2022). These evaluations make the gap plausible, though their training budgets and evaluation units differ. This study computes patient-macro balanced accuracy from patch predictions.

Imbalance damage differs strongly between the two tasks (Figure 2a). On TCGA-UT, damage is small but accelerates as ρ increases: 0.57 pp at $\rho = 10$, reaching the 1-pp threshold at $\rho = 20$ and 2.91

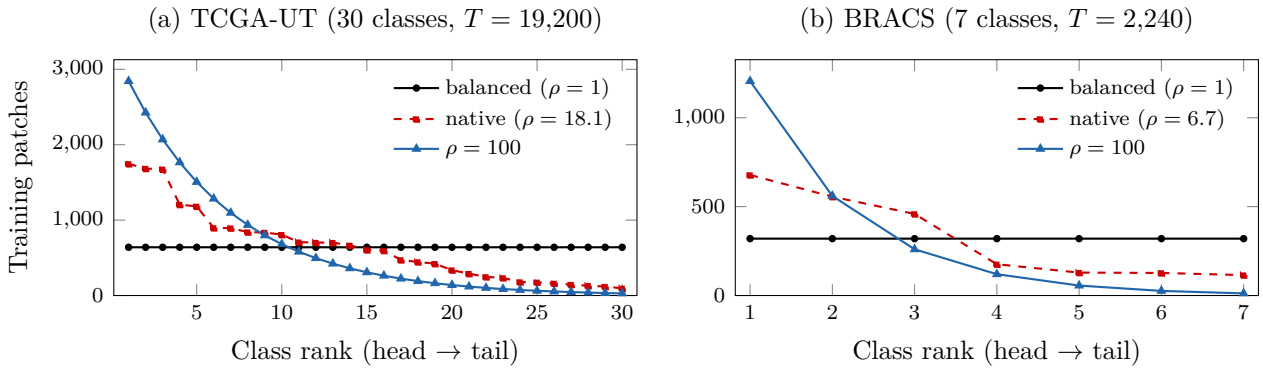


Figure 1: Training class distributions sorted from head to tail: balanced, native shares of one split, and $\rho = 100$. The tested ratios are $\rho \in \{1, 2, 5, 10, 20, 50, 100\}$; intermediate profiles lie between balanced and $\rho = 100$. Class-to-rank assignment is randomized per draw. T: total training patch budget.

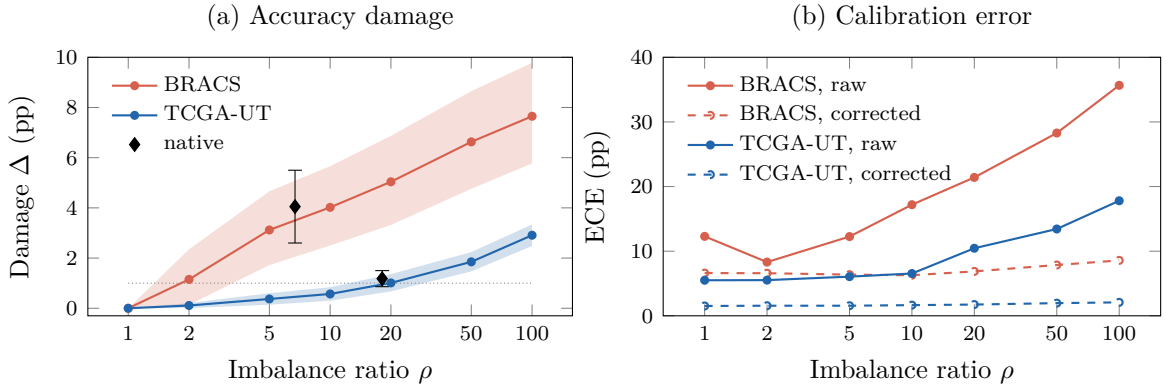


Figure 2: (a) Balanced-accuracy damage versus ρ with 95% bands; diamonds mark native class shares, dotted line the 1-pp threshold. (b) Expected calibration error (ECE) of the raw predicted probabilities (solid) and after a single global confidence correction fitted on validation data (dashed).

pp [2.48, 3.34] at $\rho = 100$. On BRACS, damage appears at $\rho = 2$ and grows steadily by about 1.15 pp per doubling of ρ: 1.15 pp at $\rho = 2$, 4.02 pp at $\rho = 10$, and 7.65 pp [5.77, 9.78] at $\rho = 100$, a fifth of the balanced accuracy. The native allocations lie on their respective curves (1.18 pp on TCGA-UT and 4.05 pp on BRACS). Aggregate damage hides a large shift in recall from tail to head. At $\rho = 100$, the TCGA-UT tail third loses 11.8 pp while the head gains 5.4 pp; on BRACS, the two tail classes drop from 38.7% to 7.3% recall.

Calibration deteriorates far more than accuracy (Figure 2b). Expected calibration error (ECE), the average gap between the model's confidence and its accuracy, rises from 5.5 to 17.8 pp on TCGA-UT and from 12.3 to 35.7 pp on BRACS at $\rho = 100$: the model becomes overconfident, but uniformly so. Temperature scaling divides all logits by one scalar fitted on validation data (Guo et al., 2017). This global confidence correction removes 92–96% of the increase.

Replacing Virchow2 with UNI2-h changes the damage little. On TCGA-UT, UNI2-h's balanced baseline is 1.78 pp higher, and its damage at $\rho = 100$ is 0.86 pp lower (97.5% interval 0.40 to 1.33): a resolved difference, but below the 1-pp threshold. On BRACS, the difference is unresolved, and tail-class recall falls to about 9% with both encoders.

4 Causes of class imbalance damage

Two questions explain the damage. The first is which changes in the training data cause it (Section 4.1). The second is why the same imbalance ratio costs more for some rare classes than for others (Section 4.2).

4.1 Sources of damage

When a tail class receives fewer training patches, two things change. Its errors contribute less often to the training loss, so common classes have more influence on the fitted decision boundary. This is the *frequency* effect: how strongly each class's available examples shape the classifier. The model also sees fewer distinct tail patches, only one or two per patient at $\rho = 100$, so it has less evidence about how that class varies. This is the *support* effect: giving those few patches more weight cannot supply the missing examples.

The conditions in Table 1 separate these effects. F keeps all patches from the balanced condition but gives head classes more weight, isolating the frequency effect. S keeps the sparse patches from the imbalanced condition but weights classes equally, isolating the support effect. The ratio condition r combines both effects. The interaction $I = \Delta_{\mathrm{r}} - \Delta_{\mathrm{F}} - \Delta_{\mathrm{S}}$ measures any additional damage when both occur together.

Class influence	Many tail patches	Few tail patches
Equal	balanced (ref.)	S: support only
Head-favoured	F: frequency only	r: ratio (both)

Table 1: Four training conditions separate the effect of unequal class influence (frequency) from the effect of fewer tail examples (support) at each ρ.

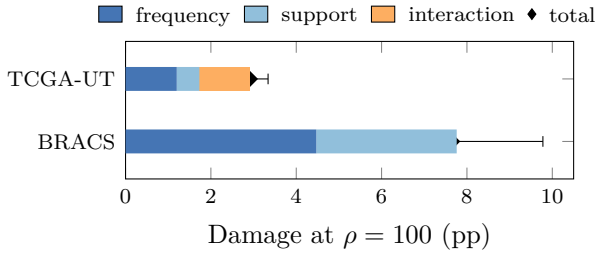


Figure 3: Damage at $\rho = 100$ by component; diamonds give the total with its 95% interval.

The damage components differ between tasks (Figure 3). On TCGA-UT, the frequency effect accounts for damage at moderate ratios (0.73 of 1.01 pp at $\rho = 20$), while sparse tail support alone costs nothing. At $\rho = 100$, the two effects compound: frequency 1.20 pp [0.87, 1.53], support 0.53 pp [0.36, 0.70], and a positive interaction of 1.18 pp [1.00, 1.37]. On BRACS, both factors are material at $\rho = 100$: frequency 4.47 pp [2.84, 6.30] and support 3.29 pp [1.99, 4.67]. Their sum matches the total damage. The interaction is near zero, −0.11 pp, but its interval [−1.74, 1.36] cannot rule out compounding of up to 1.4 pp. Sparse tail support alone costs BRACS more than total TCGA-UT imbalance damage.

4.2 Vulnerable classes

On BRACS, rotating every class through every rank shows that easy classes cost most when rare. Damage at $\rho = 100$ ranges from 5.73 pp (usual ductal hyperplasia, UDH, in the tail) to 9.99 pp (invasive carcinoma), because a class with high balanced recall has the most to lose (Figure 4a).

On TCGA-UT, a class is especially vulnerable when it receives few patches but a frequent confusion partner receives many. A score fixed before testing new class-to-rank assignments predicted their damage from this pattern. The assignment ranked most damaging costs 4.51 pp, the least damaging 1.34 pp, and random assignments 2.91 pp (Figure 4b). Reversing the most damaging assignment preserves pairwise rank gaps but lowers damage by 1.11 pp [0.47, 1.73]. Damage is therefore greatest when the class prone to confusion is the rarer one.

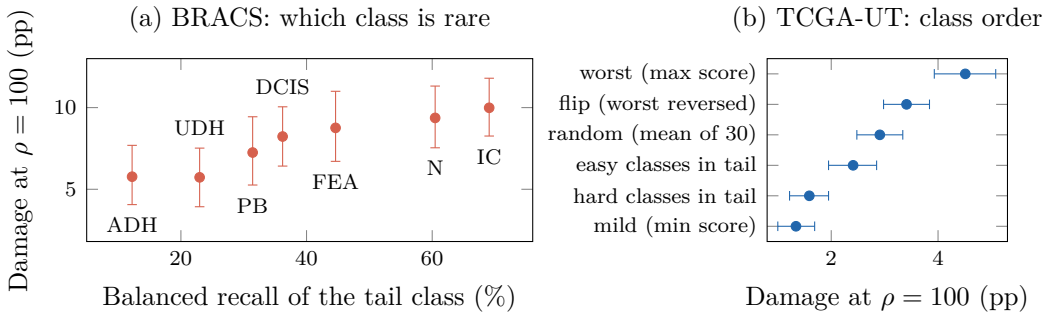


Figure 4: Damage at $\rho = 100$ with 95% intervals. (a) BRACS tail-class identity versus balanced recall (Spearman correlation 0.96, $n = 7$). ADH: atypical ductal hyperplasia; UDH: usual ductal hyperplasia; PB: pathological benign; DCIS: ductal carcinoma in situ; FEA: flat epithelial atypia; N: normal; IC: invasive carcinoma. (b) TCGA-UT class-to-rank assignments.

A Guide to the supporting experiments

The numbered folders below are under `experiments/03_class_imbalance/`. Each folder's `report/` directory contains a PDF and its LaTeX source. These reports provide the protocols, full tables, and uncertainty estimates for the results summarized here.

Headline results.

- **Imbalance curves and probability quality:** Reports 25 and 26, in `00_damage/25_damage_tcga_ut/report/` and `00_damage/26_damage_bracs/report/`, give the balanced baselines, damage at every ratio, native allocations, head-to-tail recall, and calibration before and after temperature scaling.
- **Class influence versus tail support:** Reports 27 and 28, in `01_cause/tcga_ut/27_prior_vs_support/report/` and `01_cause/bracs/28_prior_vs_support/report/`, show the paired frequency-only (called prior-only, P, in the reports) and support-only conditions, their interaction, and its uncertainty on each dataset.
- **Which classes become rare:** Report 29, in `01_cause/bracs/29_tail_assignment/report/`, rotates every BRACS class through the tail. Report 32, in `01_cause/tcga_ut/32_worst_assignment_rate/report/`, tests predicted damaging and mild class orders on TCGA-UT, including reversal of the most damaging order.
- **Encoder comparison:** Report 39, in `01_cause/cross/39_encoder_transfer/report/`, compares Virchow2 and UNI2-h on matched patients and patches and examines sensitivity to regularization selection.

Further reading. Reports 25 and 26 also track negative log-likelihood and fitted confidence corrections across the full imbalance grid. Reports 29 and 32 give class-level recall and confusion analyses behind the aggregate class-order effects, useful when the identity of a rare class matters. Report 39 adds per-class encoder comparisons and a fixed-regularization analysis, showing how the probe-selection rule affects the encoder contrast.

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